The evolution of affordability and availability in the housing market for young adults in the Netherlands

Tutorial group 1, C. Shouwenaar

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Chapter 1: Identifying the Social Problem

The Dutch housing market has changed significantly over the past two decades. For young adults, finding suitable housing has become increasingly difficult due to rising house prices, higher rents, and a continuing housing shortage (Nederlands Jeugdinstituut, n.d.). As a result, many young people struggle to enter the housing market.

This research examines how housing affordability and availability for young adults in the Netherlands changed between 2005 and 2025. It focuses on trends in house prices, rental costs, incomes and housing shortages to understand the main developments during this period.

1.1 Describing the Social Problem

The Dutch housing market has become increasingly inaccessible for young adults, making the move toward independent living far more difficult. This is socially relevant because stable and affordable housing is a foundation for financial independence, mental well-being, and equal life opportunities. When young people cannot secure a home, they delay major life steps such as studying, working full-time, or forming households—effects that ripple through society. The OECD notes that the Netherlands is among the countries where young adults face the greatest barriers to affordable housing due to rising prices and persistent shortages (OECD, 2021).

A combination of sharply increasing house prices, higher rents, and limited supply has created a structural imbalance between demand and availability. These pressures fall disproportionately on first-time buyers and young renters, widening generational inequality: older households benefit from accumulated housing wealth, while younger cohorts struggle to enter the market at all.

Chapter 2: Data Sourcing

2.1 Load in the data

2.2 Short summary of the datasets

In our research we use 6 datasets giving us valuable information about housing prices, new-built housing, population growth and inflation:

-Inflatie.csv
-Welvaart_van_huishoudens___kerncijfers.csv
-Bevolkingsgroei_opbouw_2005_2019.csv
-Huizenprijzen_Huizenvoorraad.csv
-Bestaande_koopwoningen_verkoopprijzen_woningtype.csv
-bruto_en_besteedbaar_inkomen_leeftijdsgroep_nieuw.csv

2.3 Description and Data Collection

The variables included in our project are **quantitative** in nature. They consist of continuous economic indicators—such as monetary values (income, house prices) and percentages (inflation)—as well as discrete metrics tracking physical counts (population size, housing stock). Categorical (nominal) variables such as `Provincies` and `Woningtype` are utilized to allow for spatial and structural segmentation.

Unlike survey data which relies on self-reported interviews, these datasets are sourced entirely from **administrative registries** compiled by Statistics Netherlands (CBS). Specifically, the socio-economic and income data is derived from the Dutch Tax Authority (Belastingdienst), while the housing metrics are registered through the Municipal Base Registry for Housing (BAG) and the Dutch Land Registry (Kadaster). This administrative nature ensures high data coverage and eliminates traditional interview biases such as recall or non-response bias.

2.4 Stated limitations of this data

While these administrative datasets are highly reliable, we face several limitations when merging and analyzing them to study young adults:

Inconsistent Demographic Definitions: The datasets categorize "young adults" differently. The income datasets slice data by the age of the primary wage earner (`leeftijd_kostwinner`), whereas the demographic dataset tracks individuals aged 20 to 45 (`jong_volwassenen_20_tot_45_jaar`). This structural mismatch limits our ability to perfectly isolate a single, uniform age bracket. **Temporal Truncation:** The demographic tracking for population build-up (`Bevolkingsgroei_opbouw_2005_2019.csv`) is currently limited to the 2005–2019 period within our data folder. This creates a data gap for the final years of our 2005–2025 analysis window. **Level of Aggregation:** Because the data is highly aggregated by the CBS to protect privacy, we can only join these files on the macro-level using the temporal dimension (`Jaartal/Jaren`) and the regional dimension (`Provincies`). We lack micro-data, meaning we cannot track individual household trajectories or specific cross-sections over time.

Chapter 3 - Quantifying

3.1 Data cleaning

We clean all of our datasets to make it clearer and simpler. We do this because by default some have strange and long titles that make it hard to work with. In this part we use 6 functions to clean up and/or merge each dataset.

1) rename() : to change the column titles.
2) select() : to select only the useful information.
3) group_by() : to split the datagroups based on a variable.
4) summarise() : to group the right data together.
5) left_join() : to join datasets together.
6) pivot_wider(): to widen datasets.

3.2 Describing important variables

We have many datasets and therefore many different variables. To name all of them would result in a long chaotic list. Therefore we chose to only name the ones that are important for our visualization.

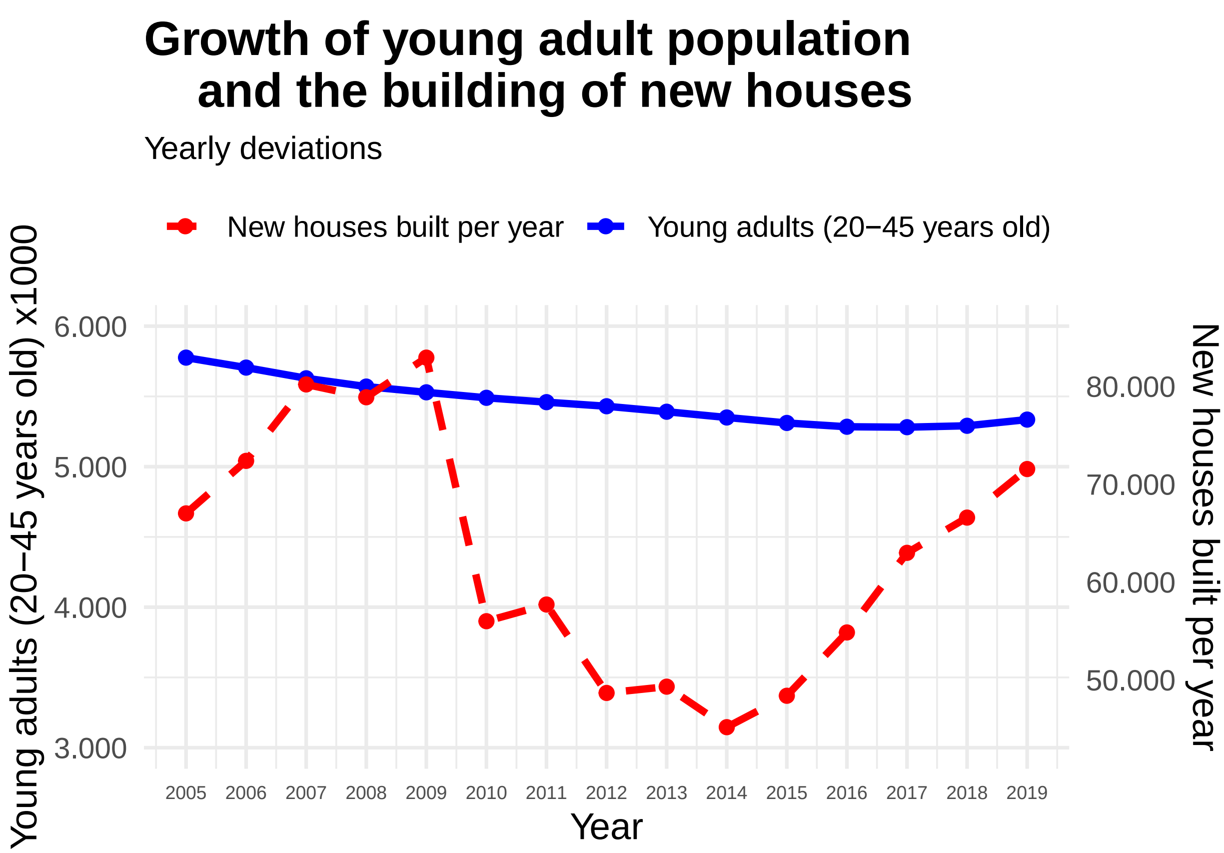
Variable	Type	Meaning
Jaartal/Jaren	Integer	Year
jong_volwassenen_20_tot_45_jaar	Integer	Number of young adults in a certain year
Huizenvoorraad	Integer	Number of new houses built in a certain year
Max_nominal_mortgage_loan_euro	Integer	Maximum nominal mortgage loan (euro)

Variable	Type	Meaning
Max_real_mortgage_loan_euro	Integer	Maximum real (inflation-adjusted) mortgage loan (euro)
Gemiddelde_verkoopprijs_euro_ Totaal_woningen	Numeric	Average sale price of all dwellings (euro)
Gemiddelde_verkoopprijs_euro_ Tussenwoning	Numeric	Average sale price of a terraced/mid-row house (euro)
Gemiddelde_verkoopprijs_euro_ Appartement	Numeric	Average sale price of an apartment (euro)
Gemiddelde_verkoopprijs_euro_ Hoekwoning	Numeric	Average sale price of a corner house (euro)
provincie	Character	Province name (with the " (PV)" suffix removed)
index	Numeric	Housing stock indexed to 2005 = 100
per_1000	Numeric	New-built homes per 1,000 existing homes

Chapter 4: Housing availability

4.1 Population growth and new houses built

Visualizing event associations



Explanation

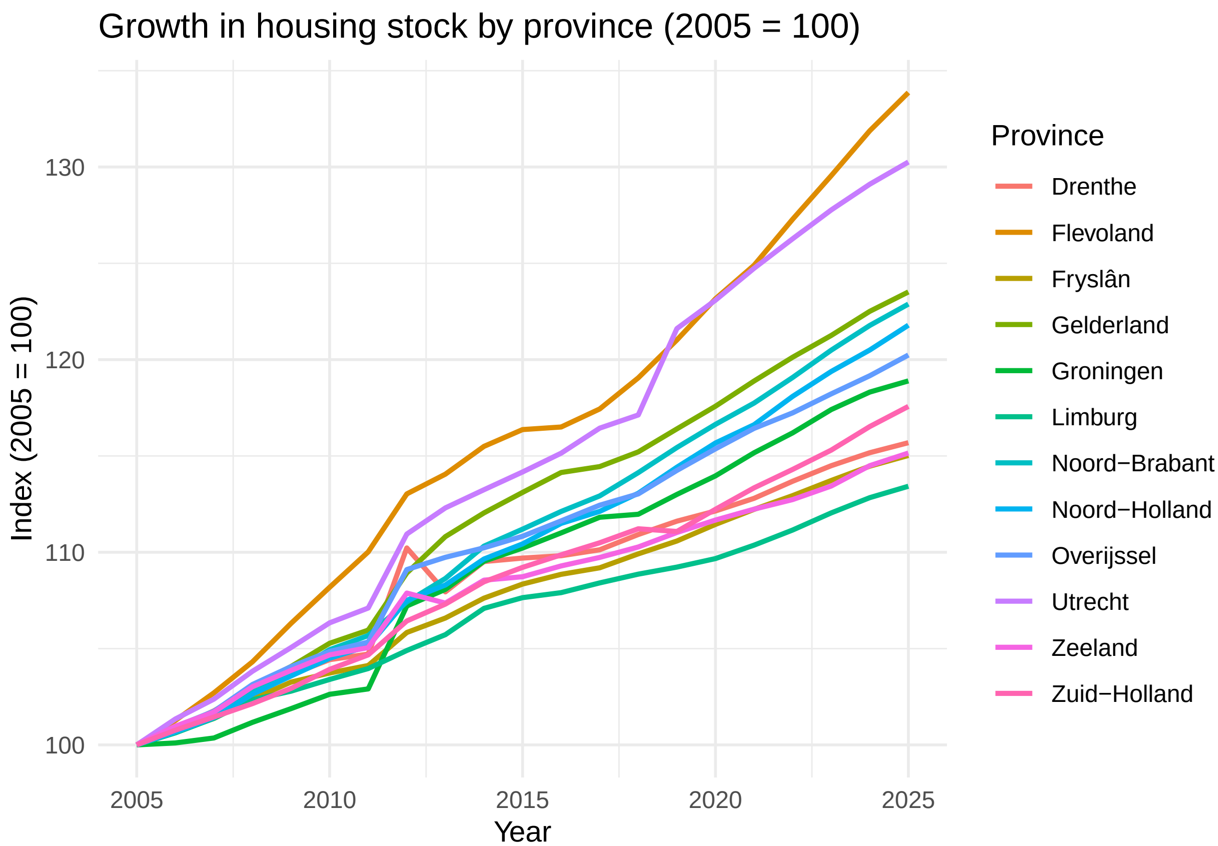
The graph compares newly built houses (yearly growth) with the total population of young adults. The data shows a slight drop in young adults across the Netherlands, a European-wide trend driven by lower birth rates (Krysztoszek, 2025). Meanwhile, housing development shifted sharply. After the 2008 global financial crisis, construction plummeted, hitting bottom in 2014 as uncertainty stalled projects (Liefting & Berkhout, 2025). Construction picks up after 2014, but not enough to make up for the lost years.

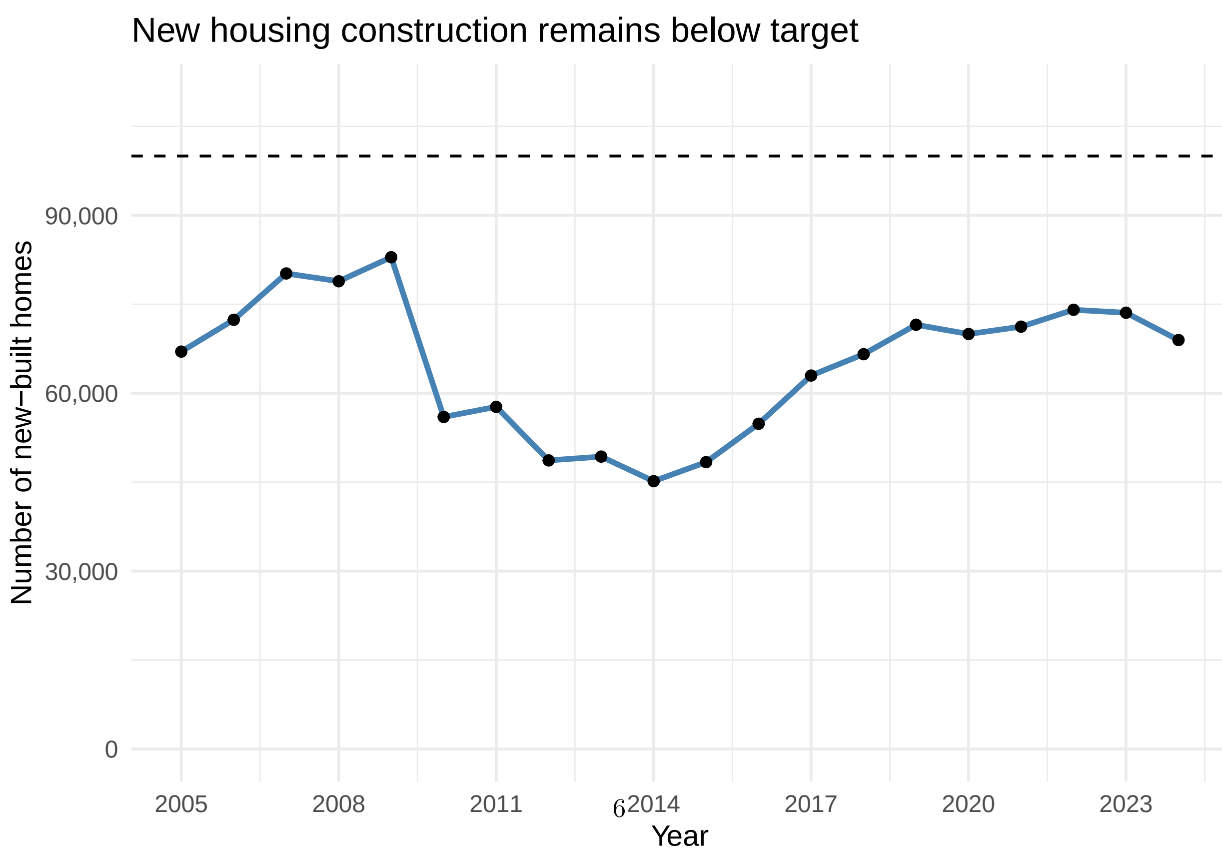
Conclusion

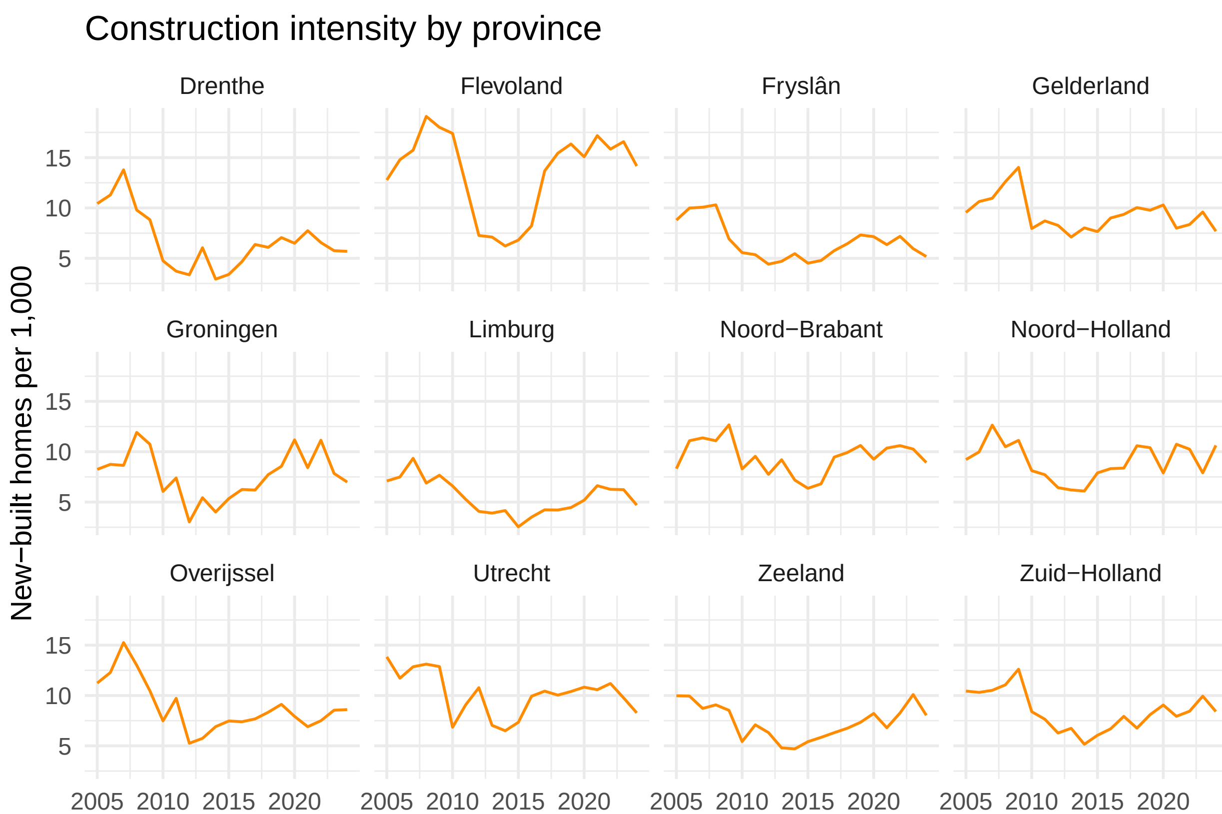
We couldn't get population data for this age group past 2019, but we expect it to stay pretty stable through 2026. Housing construction is a different story. It's getting heavily restricted right now by environmental rules, specifically the "stikstofcrisis" (Liefting & Berkhout, 2025). Because of this bottleneck, we'll likely see a big drop in new homes being built. Putting it all together, the housing shortage is caused by that dip in construction from 2009-2014 and this expected new drop after 2019, all while the population stays about the same.

4.2 Housing per province

Temporal







Explanation

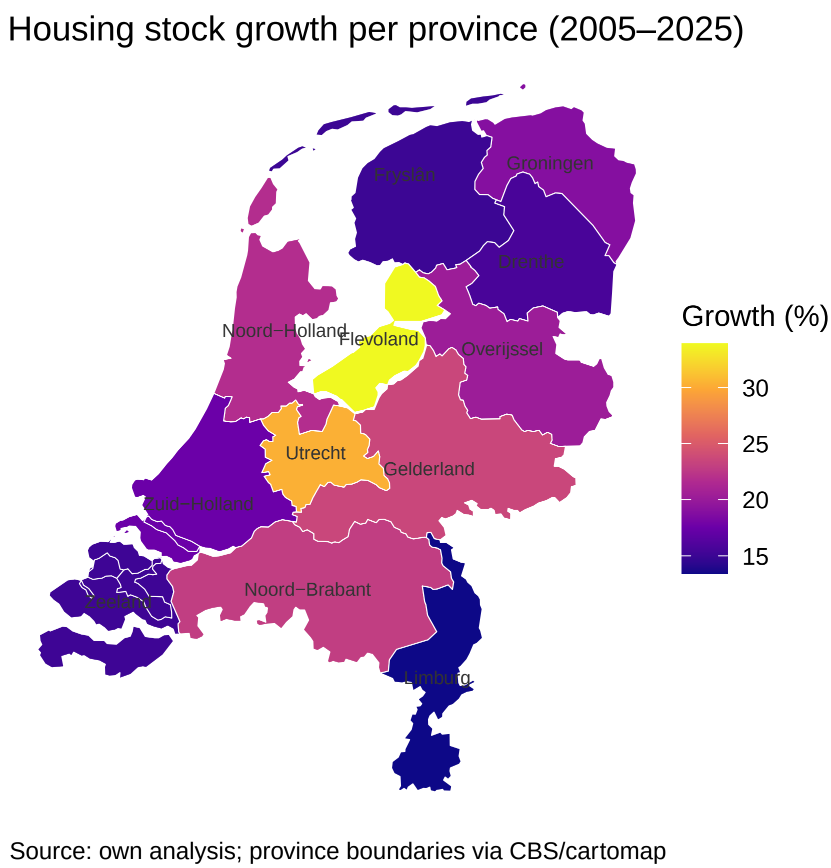
We put the housing data into three charts, because housing stock and new construction are actually two different things. The stock is the total number of homes that exist, and new construction is what gets added each year. Putting those two in one chart with the same index would be a bit misleading, since the stock rises slowly while new construction jumps up and down a lot from year to year. In the first chart we set the stock per province to 2005 = 100, which makes the provinces comparable even though some are much bigger than others. In the second chart we added up new construction nationally per year and put a line at 100,000, the number the government wants to build each year. One caveat: our figures only cover actual new construction, while that target also counts things like offices turned into homes, so the line is an indicative lower bound and not an exact match. The third chart shows building intensity (new construction per 1,000 existing homes), which corrects for how big a province is. We left 2025 out for new construction, because that value is missing from the dataset, so we don't draw conclusions about that year.

Conclusion

Over time the housing stock grew steadily in every province, but new construction tells the more interesting story. After a peak around 2009 it dropped to a low of about 45,000 homes in 2014, and only climbed back to around 74,000 by 2022, never reaching the target of 100,000. The timing of that dip lines up with the 2008 credit crisis, but this is an association and not proof: the decline only set in with a delay (building has a long pipeline), and the later recovery was also shaped by other things like low interest rates and rising demand after 2020, so the crisis alone can't explain the whole pattern. It's also important to say that our data only measures supply, not the shortage itself. The link to demand comes from outside sources: the number of households has been growing faster than the housing stock (Rijksoverheid, n.d.), which left a statistical shortage of roughly 396,000 homes in 2025 (Volkshuisvesting Nederland, 2025). We cannot show that shortage from our own data alone, but putting the two together suggests that for young adults the problem isn't only about prices, but also about availability: too few homes were added to keep up with the growing demand.

4.2 Housing

Spatial variation



Explanation

To show where in the country the growth happened, we mapped the total growth of the housing stock per province between 2005 and 2025. For each province we worked out the percentage change in the stock and used that to colour the map. The province borders come from a public CBS/cartomap file that we read in as a GeoJSON, so we didn't need any extra mapping software. One small caveat: the province labels sit at the average of each province's border points, which is only a rough centre, so for oddly shaped provinces the label can land slightly off. The colours and the actual figures aren't affected by that.

Conclusion

The map shows a clear centre–periphery pattern. Growth is highest in the central provinces Flevoland (+34%) and Utrecht (+30%) and lowest in the ones on the edges, with Limburg in the far southeast (+13%) and the coastal provinces Zeeland and Fryslân (around +15%) at the bottom. This matters for the research question, because the stock grew fastest exactly in the central regions where demand from students and young workers is highest, while the quieter provinces on the edges grew the least. So the new supply wasn't added where it would help the most, which helps explain why housing stayed hard to find for young adults in precisely the areas they most want to live.

Chapter 5 Housing affordability

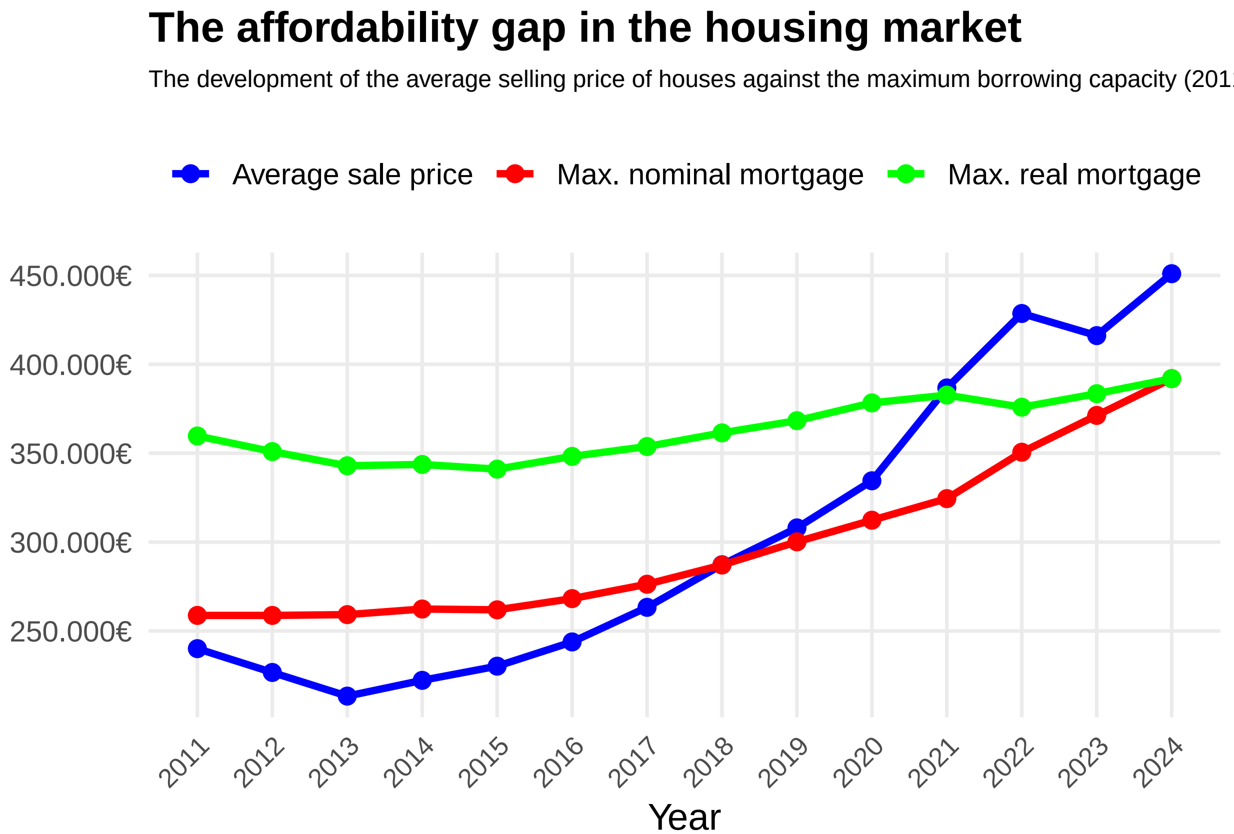
5.1 New variables

The first, necessary variable added is the maximal nominal mortgage loan that young adults can borrow from the bank. The maximal nominal mortgage loan is based on the LTI and the median gross income. The LTI (Loan-to-Income) is the ratio between the gross income and the mortgage loan. This ratio decides the maximum amount that can be borrowed, which usually is around 4.5 times the gross income in the Netherlands (Homefinance Hypotheken, n.d.). The choice to use the median gross income over the average gross income is supported by the fact that the mean usually is distorted by high extremes, while the median gives a representative reflection of the "typical" young adult.

The Netherlands has one of the highest mortgage debts of Europe, which indicates that the Dutch housing market is really dependent on credits (DNB, n.d.). Because the Dutch housing market for buyers is fundamentally credit-driven and reliant on the LTI, variables solely addressing income are not enough to model the maximum purchasing capacity of young adults. However, this nominal borrowing ceiling is the variable that can demonstrate the maximum buying capacity of young adults. Moreover, this variable can be rightfully compared to the housing prices. All in all, the maximal nominal mortgage loan more or less serves as the baseline for the study regarding housing affordability.

Another variable that is valuable, especially in comparison with the nominal maximum mortgage loan, is the real maximum mortgage loan. In order to calculate this, we set the year 2024 as the base year and set the adjustment factor to 1 in 2024. In this year, the nominal value is equal to the real value. The adjustment factor quantifies how much prices have risen with base year of 2024. The real maximum mortgage loan is adjusted for inflation and clearly proves the money illusion and the sudden inflation shocks. In order to create the extra column, it is necessary to use the cumprod() function, as prices rise exponentially.

5.2 Temporal variation: Affordability gap in the housing market



Explanation

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Conclusion

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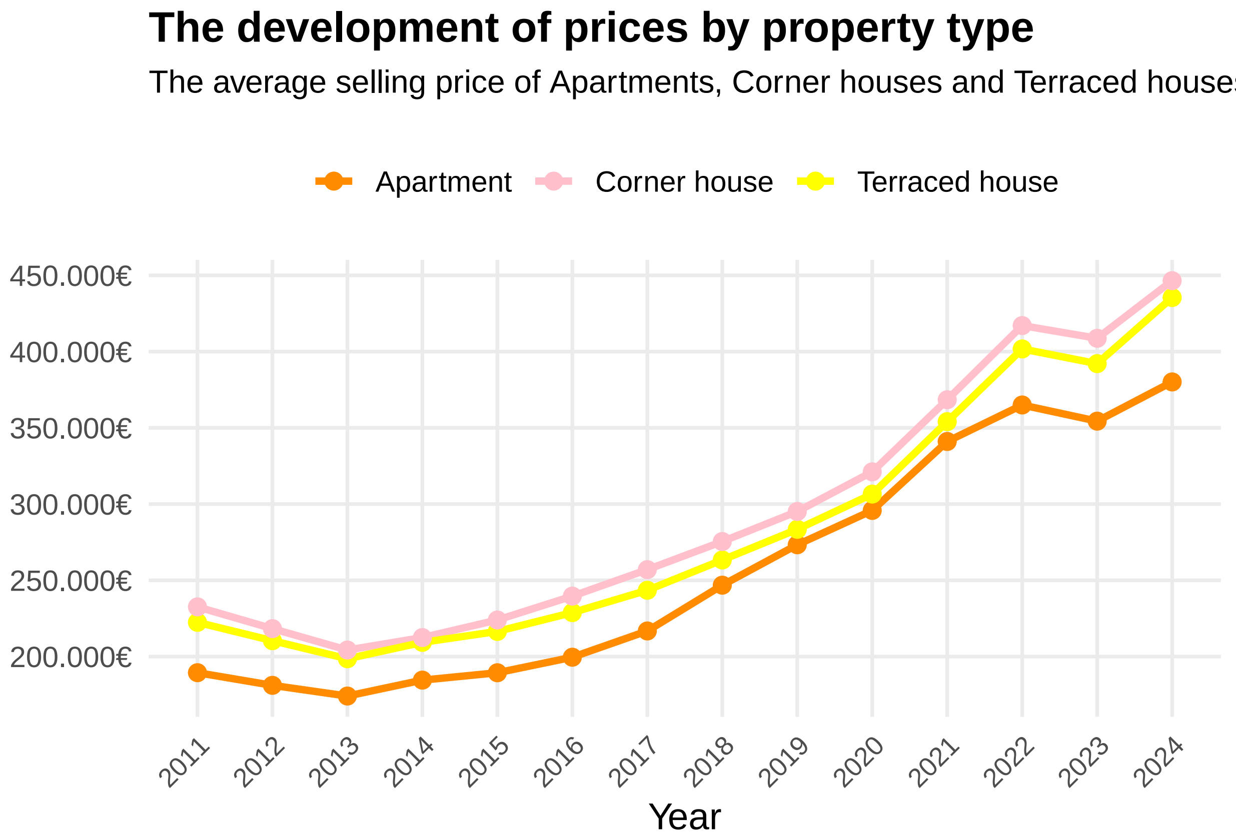
5.3 Subgroup analysis: The development of prices by property type



Explanation

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Conclusion

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Chapter 6 - Discussion

6.1 Discuss your findings

Chapter 7 - Reproducibility

7.1 Github repository link

This is the link to our public repository: https://github.com/TimKnegt/Tutorial-group-1-1

7.2 Reference list

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